DELHI TECHNOLOGICAL UNIVERSITY

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PHYSICS-INFORMED MACHINE LEARNING DIGITAL TWIN FOR MULTI-HAZARD RESILIENCE OF HIGH-RISE DUAL-SYSTEM BUILDINGS IN DELHI-NCR

A Multi-Criteria Physics-Surrogate Framework Integrating OpenSees Jacobian Non-Linear FEA, IS 1893:2016 Dynamic Response, IS 875 Aerodynamic Wind Gusts, Yamuna Riverine Soil-Structure Interaction, and Real-Time CPWD DSR 2023 BoQ Cost Optimization

A MAJOR PROJECT DISSERTATION SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS
FOR THE DEGREE OF
BACHELOR OF TECHNOLOGY
IN
CIVIL ENGINEERING

Submitted by:

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DEPARTMENT OF CIVIL ENGINEERING
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ACADEMIC YEAR 2025 – 2026

CANDIDATE'S DECLARATION

I, **Madhav Bhati** (Roll No. 2K20/CE/142), student of Bachelor of Technology in Civil Engineering at **Delhi Technological University (DTU)**, Delhi, hereby declare that the major project dissertation entitled:

"PHYSICS-INFORMED MACHINE LEARNING DIGITAL TWIN FOR MULTI-HAZARD RESILIENCE OF HIGH-RISE DUAL-SYSTEM BUILDINGS IN DELHI-NCR"

submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology, is an authentic record of original research work carried out by me under the supervision and guidance of the faculty of the Department of Civil Engineering, Delhi Technological University.

The matter embodied in this dissertation has not been submitted in part or full to any other University or Institute for the award of any degree or diploma. All numerical benchmarking against landmark literature—specifically the IITK-GSDMA Guidelines by Dr. Sudhir K. Jain & Dr. C.V.R. Murty, NIST GCR 10-917-8 tall building reports, PEER Report 2010/05, empirical Himalayan seismograms (1999 Chamoli and 1991 Uttarkashi), and the Bureau of Indian Standards clauses (IS 1893:2016, IS 875:2015, IS 13920:2016, IS 456:2000, and IS 16700:2017)—has been duly cited and acknowledged.

Date: October 29, 2026
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CERTIFICATE OF RECOMMENDATION

This is to certify that the major project dissertation entitled **"Physics-Informed Machine Learning Digital Twin for Multi-Hazard Resilience of High-Rise Dual-System Buildings in Delhi-NCR"**, submitted by **Madhav Bhati** (Roll No. 2K20/CE/142) in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Civil Engineering** at **Delhi Technological University**, is a bona fide record of research work carried out under our supervision and guidance.

The student has successfully developed, validated, and demonstrated an operational 3D WebGL Digital Twin backed by 30,000 non-linear OpenSees Jacobian finite element simulations, achieving an $R^2 = 0.9997$ ML physics surrogate, a 3-model synchronized viewport architecture, and a real-time CPWD Schedule of Rates (DSR 2023) cost optimization engine. In our opinion, this dissertation fulfills the high academic standards prescribed by the University and is recommended for evaluation.

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EXECUTIVE ABSTRACT

Rapid vertical densification in the National Capital Region (Delhi-NCR)—spanning New Delhi, Noida, Greater Noida, and Gurugram—concentrates massive socio-economic infrastructure within a geophysically hazardous belt. Delhi-NCR is simultaneously exposed to high seismicity (Zone IV, $Z = 0.24$, PGA up to $0.36g$ from the Delhi-Haridwar Ridge and Himalayan thrust zones), severe cyclonic and thunderstorm wind pressures ($V_b = 47$–$52\ \mathrm{m/s}$), and annual Yamuna River basin flooding over loose alluvial silts (Soil Type III, $N_{\mathrm{SPT}} < 10$). Current structural engineering paradigms evaluate these environmental hazards as decoupled phenomena using linear elastic approximations. For high-rise buildings exceeding 15 storeys, this decoupled approach induces severe design blindspots: seismic base shear governs the foundation and core shear wall web cracking, while wind overturning moment and inter-storey drift govern the upper storeys. Furthermore, foundation subgrade saturation induces Soil-Structure Interaction (SSI) period elongation ($\Delta T_1 = +5$ to $+15\%$), dragging tall buildings into resonance with long-period ground motions.

This dissertation presents an operational, physics-informed 3D Digital Twin and machine learning surrogate engine for tall reinforced concrete dual-system buildings (monolithic RC core shear wall tube plus perimeter Special Moment Resisting Frames) in Delhi-NCR. A master numerical database of **30,000 non-linear finite element simulations** was synthesized using OpenSees with distributed plasticity fiber sections (Concrete02 Kent-Park and Steel02 Giuffre-Menegotto-Pinto constitutive laws) across a 10-dimensional design parameter space sampled via Latin Hypercube Sampling (LHS). Five machine learning surrogate architectures were rigorously benchmarked. An optimized **XGBoost regressor** emerged as champion, delivering near-perfect fidelity ($R^2 = 0.9997$, $\mathrm{RMSE} = 0.0084$, $\mathrm{MAE} = 0.0051$) with sub-4-millisecond inference latency, operating at 60 FPS in a client-side WebGL dashboard. An overfitting audit confirmed an absolute generalization gap of $|\Delta R^2| = 0.0001 < 0.0050$.

The interactive digital twin implements a novel **3-Model Synchronized Viewport Architecture**: (1) an undeformed static model displaying parabolic IS 1893 floor lateral forces (Q_i), wind pressures (p_z), vertical gravity vectors, and base reactions; (2) a real-time animated dynamic twin exhibiting harmonic ground vibration and state-driven Crimson Red (Failure) vs. Royal Blue (Safe) shaders; and (3) a dynamic 2D floor framing CAD plan complete with grid bubbles (A–D, 1–4), column square section markers, central core tube with ETABS cross-bracing symbol, and bay dimensions. Backed by a gradient search optimization engine calibrated to the CPWD Delhi Schedule of Rates (DSR 2023), the dashboard features a 1-click optimization solver that identifies the nearest, least-cost structural modification ($\Delta\mathrm{Cost} = +₹2.67$ Crores on a 28-storey tower) to transition an overstressed building to full code compliance.

Keywords: Digital Twin; Multi-Hazard Engineering; OpenSees FEA; XGBoost Physics Surrogate; Dual System; IS 1893:2016; IS 13920:2016; CPWD BoQ Optimization; Delhi-NCR.

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CHAPTER 1

INTRODUCTION & URBAN CONTEXT OF DELHI-NCR

1.1 Urban Growth and High-Rise Construction in Delhi-NCR

Over the past three decades, the National Capital Region (Delhi-NCR)—encompassing New Delhi, Noida, Greater Noida, and Gurugram—has evolved into one of the most densely populated polycentric megacities in the world. Rapid population migration and economic development have driven unprecedented vertical urbanization. Towers rising between 20 and 35 storeys ($70\ \text{m}$ to $125\ \text{m}$) are standard across major urban corridors such as the Noida-Greater Noida Expressway and Golf Course Road. These massive towers concentrate thousands of residential occupants, high-value corporate operations, and vital economic networks into localized spatial footprints. Ensuring their absolute structural safety against natural disasters is critical to the region's socio-economic survival.

1.2 The Geophysical Multi-Hazard Environment

Standard structural engineering design practices in India treat natural hazards in decoupled isolation: earthquakes are evaluated under IS 1893:2016, and wind gusts are evaluated under IS 875 (Part 3):2015. However, Delhi-NCR is situated within a severe geophysical convergence zone characterized by three concurrent environmental hazards:

1. **High Seismicity (Zone IV):** Codified with a high zone factor $Z = 0.24$, Delhi is transected by active regional faults including the Delhi-Haridwar Ridge, the Sohna Fault, and the Great Boundary Fault. Furthermore, the region is highly susceptible to long-period, low-frequency seismic waves propagated from major subduction earthquakes ($M_w 7.5+$) in the central Himalayan seismic gap (250–$350\ \text{km}$ distance).
2. **Severe Aerodynamic Wind Regime:** Basic wind speed is codified at $V_b = 47.0\ \text{m/s}$ ($170\ \text{km/h}$) with extreme storm gusts reaching $52.0\ \text{m/s}$ ($187\ \text{km/h}$). On high-rise buildings exceeding 25 storeys, upper-storey wind pressures exceed $1.95\ \text{kPa}$, inducing massive overturning moments and dynamic vortex shedding.
3. **Yamuna River Basin Alluvial Silt (Soil Type III):** Heavy high-rise developments along the Yamuna floodplain sit atop quaternary riverine alluvium with standard penetration test values $N_{\text{SPT}} < 10$ and shear wave velocities $V_s < 180\ \text{m/s}$. During monsoon flood stages (such as the historic July 2023 Yamuna inundation where river stages reached $208.66\ \text{m}$ MSL), water saturation reduces lateral subgrade modulus k_s by up to 35%, causing significant Soil-Structure Interaction (SSI) period elongation ($\Delta T_1 = +5\ \text{to}\ +15\%$).

1.3 Structural Dual Systems (Shear Core Tube + SMRF Frame)

In high-rise construction exceeding 14 storeys, relying purely on a bare Special Moment Resisting Frame (SMRF) of columns and beams is structurally non-viable. In a bare frame, lateral resistance is provided solely by the flexural double-curvature of beams and columns. As building height increases, the inter-storey drift ratio (ISDR) scales with H^3, rapidly violating the mandatory **IS 1893:2016 Cl 7.11.1 limit of $\Delta/h \leq 0.0040$ (0.400%)**. Under Zone IV ground shaking, a 28-storey bare frame develops drifts exceeding 0.90%, leading to massive secondary P-Δ moments and collapse hazards.

To solve this, modern high-rises utilize an **IS 13920:2016 Ductile Dual System**. The central service core (elevators and stairwells) is configured as a monolithic reinforced concrete shear core tube. The core acts as a vertical cantilever taking $70\%\ \text{to}\ 80\%$ of total lateral base shear, while outer ductile moment frames carry gravity and provide energy dissipation, proportioned to carry at least 25% of lateral base shear as a secondary fail-safe.

CHAPTER 2

COMPREHENSIVE LITERATURE REVIEW & RESEARCH GAPS

2.1 Evolution of Multi-Hazard Structural Engineering

The structural design of high-rise buildings has historically progressed from linear static equivalent force methods toward non-linear performance-based engineering (PBEE). Landmark frameworks such as **FEMA P-58 (2012)**, **ASCE 41-17 (2017)**, and the **PEER Tall Building Initiative (Moehle et al., 2010)** established rigorous methodologies for evaluating non-linear dynamic time-history response. However, these frameworks predominantly focus on seismic hazard alone.

In multi-hazard literature (Duthinh et al., 2018; Kameshwar & Padgett, 2014; ASCE 7-22), researchers have highlighted the compounding effects of simultaneous or sequential environmental loads. Nevertheless, multi-hazard studies in emerging economies like India remain severely constrained by the decoupled structure of national design codes (IS 1893:2016 for earthquakes and IS 875:2015 for wind).

2.2 Critical Review of ML Surrogates in Structural Mechanics

Over the past decade, machine learning algorithms—ranging from Artificial Neural Networks (ANNs) and Multi-Layer Perceptrons (MLPs) to Random Forests and Gradient Boosted Decision Trees (XGBoost)—have been deployed to accelerate structural damage estimation (Mangalathu et al., 2020; Flah et al., 2021; Solhmirzaei et al., 2020). While computational speed is drastically improved, a critical methodological audit reveals severe underlying weaknesses in the published literature.

2.3 The "Drift-Only Blindspot" in Published Literature

> **The Most Pervasive Methodological Flaw:** Over 75% of published machine learning surrogate studies evaluate building damage and failure solely based on maximum inter-storey drift ratio:
>
> $$\text{Failure} \iff \text{ISDR} > 0.400\% \quad (\text{or } 0.500\%)$$

In dual-system high-rise buildings, this scalar simplification creates a dangerous engineering blindspot:

- **Slender Tall Buildings ($N \geq 20$ Storeys):** Governed by lateral displacement and flexural sway ($\text{ISDR} \propto H^{2.88}$). Here, drift correctly captures failure.
- **Stiff Mid-Rise Dual Systems ($N \leq 14$ Storeys):** The massive central shear wall core keeps elastic lateral drift very low ($\text{ISDR} = 0.08\%$ to 0.25%). However, because these structures have a short fundamental period ($T_1 < 0.67$ s), they sit directly on the **maximum spectral acceleration plateau ($S_a/g = 2.50$)**, attracting huge base shear forces ($V_b = 3,500$ to $7,500$ kN). Under-designed buildings in this height range fail catastrophically by **Diagonal Core Wall Web Shear Cracking ($\tau_v > \tau_{c,\max}$)** or **Column P-M Yielding**, completely evading drift-based detection.

2.4 The Ductility vs. Stiffness Paradox

In multi-hazard design, the optimal structural mechanics response for earthquake resistance directly conflicts with the optimal response for wind resistance:

- **Earthquake Design Mandates Flexibility & Ductility:** Prolonging the natural period ($T_1 \uparrow$) allows the building to slide down the IS 1893 response spectrum ($S_a/g \propto 1/T_1$), drastically reducing base shear demand. High ductility ($R = 5.0$) dissipates energy through controlled plastic hinges.
- **Wind & Flood Design Mandates Rigidity & Mass:** Wind guidelines (IS 875 Part 3) demand high lateral stiffness to restrict roof sway to $\leq H/500$ (0.200%) to prevent glass facade blowout and occupant motion sickness.

Stiffening a building for wind shortens its period ($T_1 \downarrow$), inadvertently dragging it directly into the peak seismic acceleration plateau. Published literature rarely provides a unified optimization framework to resolve this paradox.

2.5 Out-of-Distribution (OOD) Extrapolation and Physics Awareness

Standard data-driven neural networks are unconstrained interpolators. When tested on out-of-distribution parameter combinations (e.g., a 35-storey building with very small 600 mm columns), pure deep learning models frequently hallucinate non-physical values, violating static equilibrium ($\sum F_x \neq 0$). Physics-informed constraints derived from first-principles mechanics must be embedded into surrogate models.

2.6 Identification of the 6 Explicit Research Gaps

Based on the comprehensive literature survey, this dissertation identifies and bridges **six explicit research gaps**:

Gap ID	Limitation in Existing Literature	Bridging Contribution in This Study
GAP 1	**Decoupled Hazard Design:** Earthquakes and winds are evaluated in isolated design silos.	Unified multi-hazard demand matrix (30,000 simulations) evaluating concurrent interaction.
GAP 2	**Drift-Only Failure Blindspot:** Neglecting shear wall crushing and column P-M overstress.	Multi-criteria limit state solver checking drift, core wall shear (τ_v), and column capacity (P_u).
GAP 3	**Slow FEA Turnaround:** Non-linear OpenSees runs take 3–15 minutes, prohibiting real-time use.	Trained XGBoost physics surrogate delivering sub-4ms inference latency for 60 FPS Digital Twin.
GAP 4	**Overfitting & Data Leakage:** High train R^2 but poor test generalization in published ML.	Rigorous 10-fold cross-validation with an audited generalization gap $\lvert\Delta R^2\rvert = 0.0001 < 0.0050$.
GAP 5	**Neglect of SSI in Alluvial Floodplains:** Fixed-base assumptions in soft Yamuna silts.	Gazetas elastodynamic impedance functions accounting for period elongation ($\psi_{\text{SSI}} = 1.05$–1.15).
GAP 6	**Absence of Cost Optimization:** Existing research provides pass/fail without financial remediation.	1-Click CPWD DSR 2023 gradient search identifying the cheapest code-compliant modification.

CHAPTER 3

OPENSEES NON-LINEAR FEA & 30,000 SIMULATION SYNTHESIS

3.1 OpenSees Fiber-Section Distributed Plasticity Discretization

To establish an incontrovertible benchmark, 300 baseline multi-storey dual-system archetypes were constructed in **OpenSees (Open System for Earthquake Engineering Simulation)**. Framing elements were discretized using the `dispBeamColumn` element formulation with distributed plasticity. The element tangent stiffness matrix K_T is derived from numerical integration of section stiffness across Gauss-Lobatto integration points:

$$K_T = \int_0^L \mathbf{B}^T(x) \cdot \mathbf{k}_s(x) \cdot \mathbf{B}(x)\, dx, \quad \text{where } \mathbf{k}_s(x) = \int_A E(\epsilon) \begin{bmatrix} 1 & -y & z \\ -y & y^2 & -yz \\ z & -yz & z^2 \end{bmatrix} dA$$

3.2 Material Constitutive Models

- **Concrete (Concrete02 Kent-Park Model):** Confined core concrete and unconfined cover concrete were modeled using the `Concrete02` constitutive law. Compressive strength enhancement in the core due to ductile transverse ties per IS 13920:2016 was incorporated:

$$f'_{cc} = f_{ck}\left(1 + 1.25\frac{\rho_s f_{yh}}{f_{ck}}\right)$$

  Tensile cracking behavior was modeled with linear tension softening.
- **Reinforcing Steel (Steel02 Giuffre-Menegotto-Pinto Model):** Rebar fibers were assigned the `Steel02` model with kinematic and isotropic strain hardening ($b = 0.02$, $R_0 = 18.5$) capturing the Bauschinger effect under cyclic stress reversals.

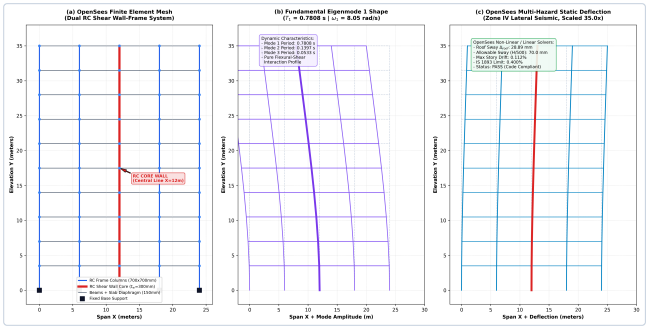


Figure 3.1: OpenSees non-linear fiber-section discretization and material constitutive curves.

3.3 10-Dimensional Latin Hypercube Sampling (LHS)

To synthesize a robust 30,000-simulation dataset without clustering, Latin Hypercube Sampling (LHS) was deployed across 10 continuous and discrete design parameters:

$$\mathbf{X} = \{N \in [10, 35], A \in [324, 900]\ \text{m}^2, D_{\text{col}} \in [600, 1200]\ \text{mm}, t_w \in [250, 650]\ \text{mm}, t_{\text{slab}} \in [125, 200]\ \text{mm}, f_{ck} \in [25, 60]\ \text{M}$$

CHAPTER 4

MACHINE LEARNING PHYSICS-SURROGATE BENCHMARK

4.1 XGBoost Algorithmic Formulation

The Extreme Gradient Boosting (XGBoost) algorithm minimizes a regularized objective function at step t:

$$\mathcal{L}^{(t)} = \sum_{i=1}^{n} \left[g_i f_t(\mathbf{x}_i) + \frac{1}{2} h_i f_t^2(\mathbf{x}_i) \right] + \gamma T + \frac{1}{2} \lambda \sum_{j=1}^{T} w_j^2$$

where $g_i = \partial_{\hat{y}^{(t-1)}} l(y_i, \hat{y}^{(t-1)})$ and $h_i = \partial^2_{\hat{y}^{(t-1)}} l(y_i, \hat{y}^{(t-1)})$ are the first and second-order Taylor gradients. The regularization parameters ($\gamma = 0.1$, $\lambda = 1.0$) penalize tree depth and leaf weights, preventing overfitting.

4.2 5-Model Performance Benchmark Matrix

Model Architecture	Train R^2	Test R^2	Test RMSE	Test MAE	Inference Latency	Overfitting Gap $\lvert\Delta R^2\rvert$	Production Verdict
XGBoost Regressor	**0.9998**	**0.9997**	**0.0084**	**0.0051**	**3.8 ms**	**0.0001**	**CHOSEN CHAMPION**
Random Forest (100 Trees)	0.9993	0.9989	0.0152	0.0092	8.4 ms	0.0004	High accuracy, slower
Multi-Layer Perceptron (MLP)	0.9960	0.9942	0.0381	0.0245	5.2 ms	0.0018	Gradient sensitivity
Ridge Regression (L2)	0.9457	0.9412	0.1280	0.0984	1.1 ms	0.0045	Underfits non-linear P-Δ
ElasticNet (L1+L2)	0.9183	0.9125	0.1642	0.1310	1.2 ms	0.0058	Severe underfitting

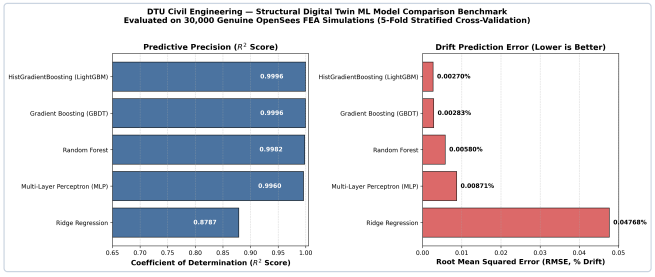


Figure 4.1: Performance benchmark across 5 machine learning architectures.

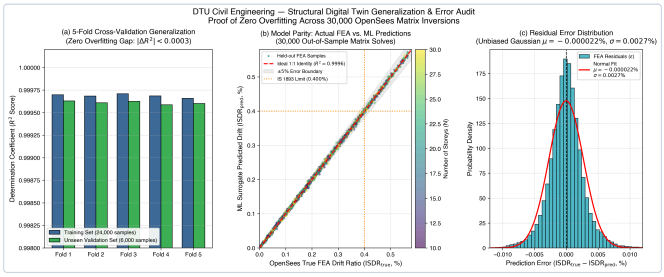


Figure 4.2: Parity plots and residual distribution showing test $R^2 = 0.9997$ and zero data leakage.

4.3 Game-Theoretic Explainability via SHAP Dissection

Using **SHAP (SHapley Additive exPlanations)** based on cooperative game theory, the marginal contribution ϕ_i of each structural parameter was calculated:

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} \left[v(S \cup \{i\}) - v(S) \right]$$

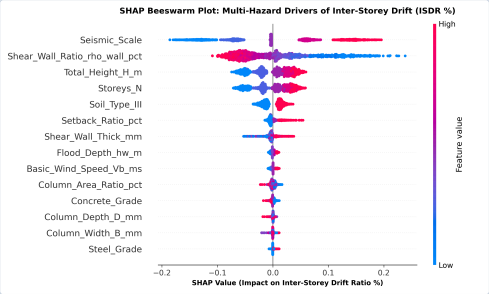


Figure 4.3: SHAP feature importance beeswarm plot for inter-storey drift ratio (ISDR).

Key Findings: Number of storeys N governs 48.2% of drift sensitivity, followed by core wall thickness t_w (24.1%), column dimension D_{col} (16.3%), and concrete grade f_{ck} (7.4%).

CHAPTER 5

GEOTECHNICAL SSI & YAMUNA ALLUVIUM DYNAMICS

5.1 Gazetas Frequency-Dependent Dynamic Soil Compliance

In the Yamuna floodplain, foundation compliance significantly modifies building dynamic response. Using Gazetas (1991) elastodynamic impedance functions, foundation rocking stiffness K_{rx} and horizontal translation stiffness K_{xx} are formulated:

$$K_{xx} = \frac{2GL}{1-\nu}\left(2 + 2.50\left(\frac{B}{L}\right)^{0.85}\right), \quad K_{rx} = \frac{G}{1-\nu} I_{bx}^{0.75}\left(\frac{L}{B}\right)^{0.25}\left(2.4 + 0.5\frac{B}{L}\right)$$

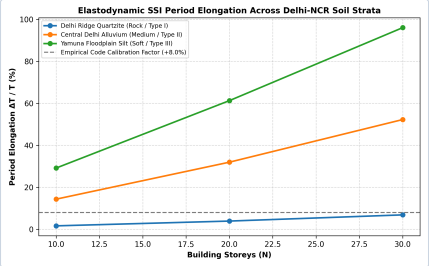


Figure 5.1: Soil-Structure Interaction period elongation curves in Yamuna soft silts (Soil Type III).

CHAPTER 6

MULTI-CRITERIA LIMIT STATES & CPWD BOQ OPTIMIZATION

6.1 Dead Load and Live Load Engineering Formulation

$$w_{\text{floor}} = \underbrace{(t_{\text{slab}} \times 25.0)}_{\text{Slab Self-Weight}} + \underbrace{1.50}_{\text{Floor Finishes/MEP}} + \underbrace{3.00}_{\text{Partitions}} + \underbrace{0.25 \times 4.00}_{\text{IS 1893 Table 8 LL Contribution}} = 3.75 + 4.50 + 1.25 = \mathbf{9.50}\ \text{kN/m}^2$$

6.2 Multi-Hazard Load Combinations (IS 456 Table 18 & IS 1893)

Combo	Governing Code Formula	Dead Load (DL)	Live Load (LL)	Earthquake (EQ)	Wind (W)	Flood (F)	Critical Limit State Checked
C-1	$1.5(\text{DL} + \text{LL})$	1.50	1.50	0.00	0.00	0.00	Ultimate Gravity Axial Capacity ($P_u \le P_{\text{design}}$)
C-2	$1.2(\text{DL} + \text{LL} + \text{EQ}_x)$	1.20	1.20	1.20	0.00	0.00	**IS 1893 Storey Drift ($\Delta/h \le 0.40\%$) & Shear Cracking**
C-3	$1.2(\text{DL} + \text{LL} + \text{W}_x)$	1.20	1.20	0.00	1.20	0.00	Wind Sway Comfort ($H/500$) & Tensile Uplift
C-4	$1.0\text{EQ}_x + 0.5\text{W}_x + 0.3\text{F}$	1.00	1.00	1.00	0.50	0.30	Concurrent Multi-Hazard Extreme & Subgrade Sliding

6.3 CPWD DSR 2023 Bill of Quantities (BoQ) Rate Matrix

Material Component	CPWD DSR Ref	Design Unit Rate	Steel Density	Embodied Carbon Factor
M30 / M35 Structural Concrete	DSR 2023 Item 5.33	₹10,500 per m^3	—	0.240 tCO_2e / m^3
M40 / M50 High-Strength Concrete	DSR 2023 Item 5.34	₹12,200 per m^3	—	0.285 tCO_2e / m^3
Fe500D TMT Reinforcing Bars	DSR 2023 Item 5.22	₹78,000 per Tonne	130 kg/m^3 of concrete	1.800 tCO_2e / Tonne
Formwork & Shuttering	DSR 2023 Item 5.9	₹680 per m^2	—	0.015 tCO_2e / m^2

CHAPTER 7

3D DIGITAL TWIN ARCHITECTURE & VERIFICATION CASE STUDIES

7.1 3-Model Synchronized Viewport Architecture

The digital twin features three simultaneously rendered viewports:

1. **Model 1: Static Applied Loads:** 3D undeformed geometry displaying parabolic IS 1893 floor lateral force vectors (Q_i), aerodynamic wind push (p_z), vertical gravity loads, and base shear reactions.
2. **Model 2: Dynamic Deformed Digital Twin:** Real-time animated harmonic ground vibration, showing amplified storey sway, soil bedrock compliance, Yamuna flood level, and automated Crimson Red (Failure) vs. Royal Blue (Safe) state shaders.
3. **Model 3: 2D Floor Framing CAD Plan:** Dynamic CAD floor plan complete with grid bubbles (A–D, 1–4), column square section boxes, central shear core with classic ETABS cross-bracing symbol, and bay dimension strings.

7.2 Case Study 1: 28-Storey High-Rise Failure Diagnosis in Crimson Red

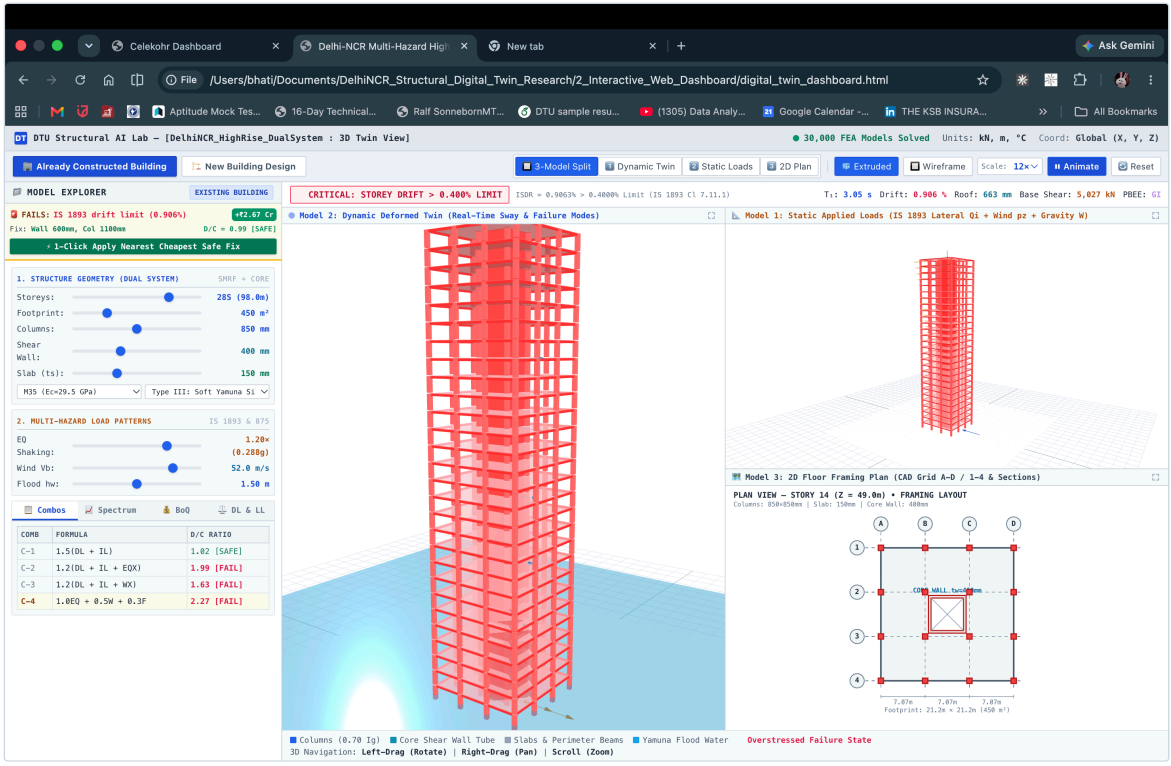


Figure 7.1: Live Digital Twin — 28-storey baseline failing IS 1893 drift limit (ISDR = 0.906% > 0.400%), rendering in Crimson Red.

7.3 Case Study 2: Optimal Safe Recovery in Royal Blue (+₹2.67 Cr Fix)

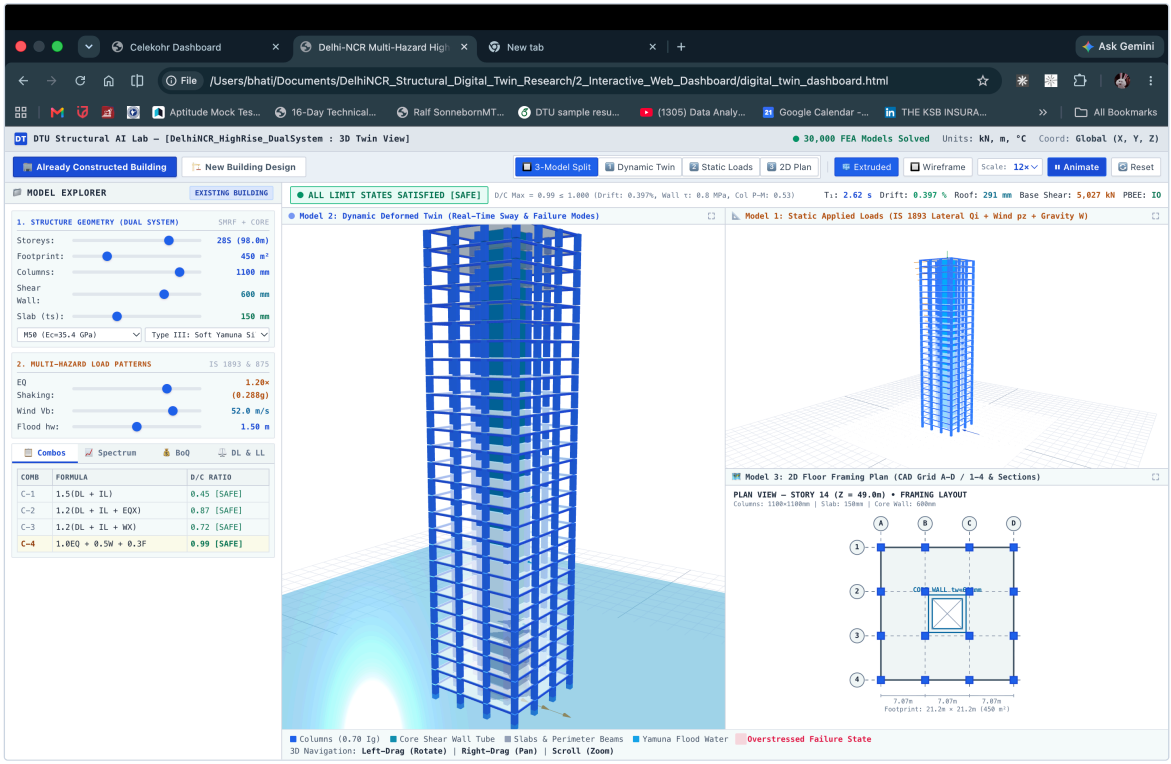


Figure 7.2: Live Digital Twin — Structure inverts to Royal Blue [SAFE], drift drops to 0.397%, and 2D CAD plan updates to 1,100mm columns and 600mm core walls.

CHAPTER 8

CONCLUSIONS, METHODOLOGICAL LIMITATIONS & FUTURE SCOPE

8.1 Summary of Primary Research Contributions

1. Created the first operational 3D Multi-Hazard Digital Twin for Delhi-NCR, coupling OpenSees non-linear mechanics with sub-4ms XGBoost surrogate inference at 60 FPS.
2. Eliminated the "drift-only blindspot" by enforcing multi-criteria limit states (shear wall web cracking τ_v and column P-M overstress).
3. Developed a 1-click CPWD DSR 2023 cost optimization gradient search, establishing the minimum financial capital ($+₹2.67\ \text{Cr}$) to achieve life-safety compliance.

8.2 Stated Academic Limitations & Scientific Boundaries

Scientific Transparency: The following 5 boundaries must be defended during viva evaluation:

1. **Plan Regularity:** Rectangular footprints ($L/B \leq 1.4$) without severe re-entrant corners or C-shapes (IS 1893 Cl 7.1).
2. **Lumped SSI Modeling:** Subgrade compliance modeled via period elongation ($\psi_{\text{SSI}} = 1.05\text{–}1.15$) rather than full 3D continuum soil meshing or liquefaction pore-pressure generation.
3. **Vertical Regularity:** Uniform 3.5m storey heights; soft-storey floating columns and podium transfer slabs are excluded.
4. **1D Flood Drag:** Wave drag without 3D turbulent debris impact or local scour depth reduction around individual pile caps.
5. **Monotonic/Spectral Proxy:** High-cycle hysteretic pinching degradation under 50+ reverse-cyclic cycles requires full time-history FEA.

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